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N-L-P

CIA -1

1) a) Regular expression for Appointment :-

```
b (0 [1-9] | [12] [0-9] / 3 [01]) / (0 [1-9] | [0-2]) / [0-9]{4} | b
```

Explanation:

b (0 [1-9] = Days from 01 - 09

[12] [0-9] = Days from 10 - 29

3 [01] = Days from 30 - 3

0 [1-9] [10-2] = Days from 1-9 to 10-2

[0-9] {4} = Exactly four digit for the year

b) Regular Expression for Indian mobile number :-

```
(?<!\d)(?:(?:) +91 [-]|0)? [6-9] [0-9] ?9) (?!\d)
```

+91 [-] = country code

0 = optional leading zero

[6-9] = from digits must be between 6 to 9

(?<!\d) and (?!\d) = prevent matching part of longer number.

c) Regular expression for a sample

(A-Z a-z 0-9 _ %.+ -]+ @ (A-Z a-z 0-9. -]+ \. (a-z A-Z]{2,}

[A-Za-z 0-9._%+-]+ = username

@ = Required separator

(A-Za-z 0-9.-]+ = Domain name

\. = Dot before the extension

d)

2) a) Finite state Automation for english noun

Lexicon: 1) Regular noun : fox, cat, dog

2) Irregular singular nouns: goose, mouse

3) Irregular plural nouns: geese, mice

4) Regular plural suffix

States:

q_0 = Start state

q_1 = Regular singular noun

q_2 = Irregular singular noun

q_3 = Regular plural noun

q_4 = Irregular plural noun

q_5 = singular noun accepted

Transitions :-

From state	input	to state
q_0	fox, cat, dog	q_1
q_1	-s	q_2
q_0	goose, mouse	q_3
q_2	end	q_5
q_0	geese, mice	q_4

5) Acceptance & morpheme sequence

sequence	Accepted	Reason
dog + -s	Yes	Regular plural
mouse + -s	NO	Irregular plural
geese	Yes	Irregular plural
fox + -s	Yes	Regular plural
mice + -s	NO	already plural

d) Inflectional vs derivational morphology

Word	type	Justification
walked	inflectional	adds -ed
unhappiness	derivational	un- and -ness
computerization	derivational	-ize and -ation
geese	inflectional	Irregular plural form.